

Supplementary Information

Helium gas or liquid nitrogen for the first thermal shield of a 4 K cryostat: conjugate heat-transfer budget, coolant operating maps and work ledger

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All cases, scripts and tabulated results: github.com/Imerson/cryostat-first-shield-He-vs-LN2.

S1 Enclosure model

Table S1. Regions, patches and boundary conditions of the enclosure CHT cases.

Region	Patch	T condition	Radiation condition
domain0 (OVC-shield gap)	warmPlate, OVC	fixed T_{hot}	grey diffuse, $\varepsilon = 0.04$
	shield_slave	fixed T_{cold}	grey diffuse, $\varepsilon = 0.02$
domain1 (inner gap)	warmPlate	fixed T_{hot}	grey diffuse, $\varepsilon = 0.04$
	coldPlate, shield_bottom	fixed T_{cold}	grey diffuse, $\varepsilon = 0.04$
	shield	fixed T_{cold}	grey diffuse, $\varepsilon = 0.02$
	domain1_to_coax_Lk	conjugate (mapped)	grey diffuse, $\varepsilon = 0.02$
coax_L1..L4 (solids)	warm cap / cold cap	fixed T_{hot} / T_{cold}	–
	lateral	conjugate to domain1	–

Table S2. Gas-region settings that make the gaps behave as vacuum. Radiation in OpenFOAM’s `viewFactor` model is solved in fluid regions, so each gap is a fluid region made inert.

Setting	Value
pressure $p = p_{\text{rgh}}$	10^{-3} Pa (<code>pRefValue</code> 10^{-3})
gravity	(0, 0, 0)
transport	constant, $\mu = 10^{-11}$ Pa s, $\text{Pr} = 0.67$
density limits	$\rho_{\text{max}} = 10^{-4}$, $\rho_{\text{min}} = 10^{-12}$ kg m $^{-3}$
radiation solver frequency	every 10 iterations
agglomeration	<code>nFacesInCoarsestLevel</code> 250, <code>featureAngle</code> 20, <code>mergeLevels</code> 1

Table S3. Harness representation. Cable UT-085-SS-SS: 304 stainless outer conductor $A = 1.585 \text{ mm}^2$, PTFE dielectric $A = 2.001 \text{ mm}^2$, 304 stainless centre conductor $A = 0.205 \text{ mm}^2$. Material $k(T)$ from the NIST cryogenic fits $\log_{10} k = \sum_n a_n (\log_{10} T)^n$. Each modelled cylinder carries n_k lines with $\kappa_{\text{eff}}(T) = (n_k/A_{\text{model}}) \sum_i A_i k_i(T)$. A_{model} is the conduction cross-section of the meshed cylinder in each stage span (calibrated from the mesh so that $(A_{\text{model}}/L) \int \kappa_{\text{eff}} dT$ equals the physical cable integral $n_k L^{-1} \sum_i A_i \int k_i dT$); it differs between the stage-1 and stage-2 geometries. Because A_{model} cancels in that product, the stage-integrated conduction is fixed by the cable data by construction: the resolved model reproduces the targets below to 0.01 %, which verifies the implementation but is not an independent prediction.

Span	$A_{\text{model}} [\text{m}^2]$	$L [\text{m}]$	L1 drive, $n_k=25$	L2 flux, 25	L3 readout, 10	L4 pump, 5
300→50 K	2.975×10^{-4}	0.335	395.8 mW	395.8 mW	158.3 mW	79.2 mW
300→77 K	2.976×10^{-4}	0.335	370.0 mW	370.0 mW	148.0 mW	74.0 mW
50→4 K	3.934×10^{-3}	0.245	27.03 mW	27.03 mW	10.81 mW	5.41 mW
77→4 K	2.845×10^{-3}	0.245	62.33 mW	62.33 mW	24.93 mW	12.47 mW

Per-bundle targets $n_k L^{-1} \sum_i A_i \int k_i dT$; shares 38.5 / 38.5 / 15.4 / 7.7 % follow from n_k with identical cable construction and length.

Table S4. Deployed $\kappa_{\text{eff}}(T) = \sum_{n=0}^5 c_n T^n [\text{W m}^{-1} \text{K}^{-1}]$, T in K] per bundle and stage span (`kappaCoeffs<8>`, last two coefficients zero). Fit error < 0.5 % over each span.

Span	Bundle	c_0	c_1	c_2	c_3	c_4	c_5
300→50 K	L1	-2.1522×10^{-1}	3.1627×10^{-2}	-2.3184×10^{-4}	9.9921×10^{-7}	-2.1908×10^{-9}	1.9383×10^{-12}
	L2	-2.1519×10^{-1}	3.1622×10^{-2}	-2.3181×10^{-4}	9.9907×10^{-7}	-2.1905×10^{-9}	1.9380×10^{-12}
	L3	-8.6092×10^{-2}	1.2651×10^{-2}	-9.2741×10^{-5}	3.9970×10^{-7}	-8.7636×10^{-10}	7.7535×10^{-13}
	L4	-4.3044×10^{-2}	6.3253×10^{-3}	-4.6369×10^{-5}	1.9984×10^{-7}	-4.3816×10^{-10}	3.8766×10^{-13}
300→77 K	L1	-1.3459×10^{-1}	2.9103×10^{-2}	-2.0205×10^{-4}	8.3171×10^{-7}	-1.7398×10^{-9}	1.4706×10^{-12}
	L2	-1.3457×10^{-1}	2.9099×10^{-2}	-2.0202×10^{-4}	8.3159×10^{-7}	-1.7396×10^{-9}	1.4704×10^{-12}
	L3	-5.3837×10^{-2}	1.1642×10^{-2}	-8.0824×10^{-5}	3.3269×10^{-7}	-6.9596×10^{-10}	5.8824×10^{-13}
	L4	-2.6917×10^{-2}	5.8206×10^{-3}	-4.0410×10^{-5}	1.6634×10^{-7}	-3.4796×10^{-10}	2.9411×10^{-13}
50→4 K	L1	-4.8624×10^{-4}	9.1378×10^{-4}	3.6621×10^{-5}	-9.0343×10^{-7}	8.4798×10^{-9}	-2.8699×10^{-11}
	L2	-4.8624×10^{-4}	9.1378×10^{-4}	3.6621×10^{-5}	-9.0343×10^{-7}	8.4798×10^{-9}	-2.8699×10^{-11}
	L3	-1.9450×10^{-4}	3.6551×10^{-4}	1.4648×10^{-5}	-3.6137×10^{-7}	3.3919×10^{-9}	-1.1480×10^{-11}
	L4	-9.7247×10^{-5}	1.8275×10^{-4}	7.3242×10^{-6}	-1.8068×10^{-7}	1.6959×10^{-9}	-5.7398×10^{-12}
77→4 K	L1	-7.4665×10^{-4}	1.2814×10^{-3}	4.9472×10^{-5}	-1.2245×10^{-6}	1.1714×10^{-8}	-4.2617×10^{-11}
	L2	-7.4665×10^{-4}	1.2814×10^{-3}	4.9472×10^{-5}	-1.2245×10^{-6}	1.1714×10^{-8}	-4.2617×10^{-11}
	L3	-2.9865×10^{-4}	5.1254×10^{-4}	1.9788×10^{-5}	-4.8978×10^{-7}	4.6855×10^{-9}	-1.7046×10^{-11}
	L4	-1.4932×10^{-4}	2.5626×10^{-4}	9.8938×10^{-6}	-2.4488×10^{-7}	2.3427×10^{-9}	-8.5230×10^{-12}

Free-molecular gas conduction bound. At $p < 10^{-2} \text{ Pa}$ the mean free path exceeds the gap and gas heat transport is free-molecular, $\dot{Q}_{\text{gas}} = \alpha \Lambda p A \Delta T$ [1], with $\Lambda = \frac{\gamma+1}{\gamma-1} \sqrt{R/(8\pi MT)}$ evaluated at the warm wall (helium, $\gamma = 5/3$: $\Lambda = 2.10 \text{ W m}^{-2} \text{K}^{-1} \text{Pa}^{-1}$; nitrogen: 1.19), accommodation coefficient α , exchange area A and $\Delta T = 250 \text{ K}$. For the enclosure's equivalent plate area $A = 1.14 \text{ m}^2$ and helium with $\alpha = 0.4$ the bound is 2.4 mW at 10^{-5} Pa , 24 mW at 10^{-4} Pa and 0.24 W at 10^{-3} Pa , that is 0.2, 2.3 and 23 % of the 1.03 W harness conduction ($\alpha = 1$: 0.6, 5.8 and 58 %). The model therefore applies to a cryostat pumped to 10^{-4} Pa or better, where the residual gas is a budget line of a few per cent; the nominal $p = 10^{-3} \text{ Pa}$ of the gap regions is a solver setting for the radiation-only vacuum, not an operating pressure.

Table S5. View-factor agglomeration refinement and row normalisation. Closure = mean row sum $\sum_j F_{ij}$ of the generated matrix (unity for a closed enclosure); loads with the matrix as generated and, at the production level, with rows normalised.

Case	Coarsest-level faces	Closure (mean; range)	Radiation, as generated		Radiation, normalised
300→50 K	60	–	9.46 W	–	
300→50 K	250 (production)	0.99–1.01; 0.50–1.13	8.41 W	7.36 W	
300→50 K	400	0.98–0.99	6.49 W	7.32 W	
300→50 K	800	0.90–0.96	3.26 W	7.27 W	
300→50 K	closed form	–	7.07 W		
300→77 K	250 (production)	0.99	8.38 W	7.34 W (closed form 7.04)	
50→4 K	250 (production)	1.00	5.29 mW	3.16 mW (closed form 3.15)	
77→4 K	250 (production)	1.00	29.7 mW	17.8 mW (closed form 17.7)	

Conservation and reciprocity audit. Two residuals are reported for every view-factor matrix. Global conservation: the net radiative power $\sum_{\text{patches}} \int q_r dA$ summed over all surfaces of a closed region, which must vanish; the generated matrices leave 1.5 W (400 faces) and 14 W (800 faces) unbalanced on the 300→50 K case, the row-normalised ones 0.10, 0.13 and 0.36 W at 250, 400 and 800 faces. Reciprocity: with the coarse-face areas A_i rebuilt from the mesh by summing the fine faces of each agglomeration (`scripts/reciprocity.py`), the area-weighted error $\sum |A_i F_{ij} - A_j F_{ji}| / \sum A_i F_{ij}$ of the normalised matrices is 3.9, 5.5 and 12.6 % at 250, 400 and 800 faces (outer gap region). Row normalisation restores closure exactly but leaves reciprocity untouched, and both residuals grow with agglomeration refinement, which is why the production level is the coarsest, 250 faces, and why the 3–4 % departure from the closed form is quoted as the accuracy of the method.

Row normalisation. Each fluid region of the enclosure model is a closed surface, so every row of the view-factor matrix must sum to unity. The generated matrix (`viewFactorsGen`, Gauss quadrature tolerance 0.01) is stored sparsely, one entry per visible face pair; the normalisation divides each row by its sum and leaves rows with a sum below 0.2 (none occurred) untouched. The solve was then restarted from the converged production field for 600 further iterations. Script `normalise_F.py` in the repository.

Table S6. Emissivity sensitivity (production view factors) and MLI-equivalent variant; the MLI-equivalent cases are also given with the row-normalised matrix, which is the value carried into the main text and the ledger.

Case	ε warm / cold	Radiation [W]	Total [W]
300→50 K, low	0.02 / 0.01	5.45	6.48
300→50 K, baseline	0.04 / 0.02	8.41	9.44
300→50 K, high	0.06 / 0.03	11.48	12.51
300→77 K, low	0.02 / 0.01	5.44	6.40
300→77 K, baseline	0.04 / 0.02	8.38	9.35
300→77 K, high	0.06 / 0.03	11.44	12.40
300→50 K, MLI-eq.	0.04 / 0.003	1.81	2.84
300→77 K, MLI-eq.	0.04 / 0.003	1.81	2.77
300→50 K, MLI-eq., rows normalised	0.04 / 0.003	1.69	2.72
300→77 K, MLI-eq., rows normalised	0.04 / 0.003	1.68	2.65

S2 Cold-plate sweep: full results

Table S7. All cold-plate runs at 9.4 W (production mesh unless marked). $\dot{m}$ in g s^{-1} , volumetric flow in $\text{cm}^3 \text{s}^{-1}$, Δp in Pa, $\dot{W}_p$ in μW , temperatures in K, h in $\text{W m}^{-2}\text{K}^{-1}$; $T_{\text{in}} = 45 \text{ K}$ (He), 77 K (LN_2). T_w is the area mean over the heated-wall patch, $T_{w,\text{max}}$ and $T_{w,\text{min}}$ its extreme face values (`surfaceFieldValue` on the patch); $\text{margin} = T_{\text{sat}}(5 \text{ bar}) - T_{w,\text{max}} = 94.0 \text{ K} - T_{w,\text{max}}$; $y^+ =$ maximum on the heated wall; $\text{closure} =$ energy-balance error. Gravity-on (g on) nitrogen runs did not reach a steady state (see closure).

Case	Re	$\dot{m}$	$\dot{V}$	Δp	$\dot{W}_p$	T_{out}	T_w	$T_{w,\text{max}}$	$T_{w,\text{min}}$	$T_w - T_b$	h	Nu	f	NTU	ε	Gr/ Re^2	margin	y^+	closure %
He 1 bar $Re=500$	500	0.159	167	0.294	48.6	56.36	87.4	106.1	47.9	36.72	17.1	3.66	0.186	0.31	0.266	0.63	—	—	-0.06
He 1 bar $Re=500$ (g on)	500	0.159	167	0.331	54.6	56.21	67.5	95.2	47.9	16.87	37.1	7.96	0.21	0.674	0.49	0.29	—	—	-1.41
He 1 bar $Re=500$ (variable properties)	500	0.159	165	0.309	51.1	56.37	83.0	97.3	48.0	32.29	19.4	4.16	0.196	0.352	0.297	0.63	—	—	-0.01
He 1 bar $Re=1000$	1000	0.318	316	0.671	211	50.68	76.8	92.6	47.1	28.94	21.7	4.64	0.112	0.196	0.178	0.15	—	—	-0.05
He 1 bar $Re=1500$	1500	0.477	465	1.12	521	48.79	71.7	85.9	46.8	24.77	25.3	5.42	0.0847	0.153	0.142	0.06	—	—	-0.05
He 1 bar $Re=2300$	2300	0.731	702	1.98	1.39e+03	47.47	67.1	79.7	46.6	20.82	30.1	6.45	0.0642	0.119	0.112	0.022	—	—	-0.06
He 1 bar $Re=2300$ (g on)	2300	0.731	703	2.02	1.42e+03	47.48	62.7	80.9	46.6	16.41	38.2	8.18	0.0656	0.151	0.14	0.018	—	—	0.55
He 1 bar $Re=2300$ (coarse grid)	2300	0.731	702	1.95	1.37e+03	47.47	67.1	79.9	47.3	20.85	30.1	6.44	0.0632	0.119	0.112	0.022	—	—	-0.03
He 1 bar $Re=2300$ (fine grid)	2300	0.731	702	1.99	1.4e+03	47.47	67.0	79.7	46.2	20.79	30.1	6.46	0.0647	0.119	0.112	0.022	—	—	-0.07
He 1 bar $Re=2300$ (variable properties)	2300	0.731	702	1.98	1.39e+03	47.47	66.0	76.9	46.7	19.80	31.7	6.78	0.0643	0.125	0.117	0.022	—	—	-0.01
He 1 bar $Re=5000$	5000	1.59	1.5e+03	6.62	9.95e+03	46.14	55.3	61.3	46.5	9.73	64.4	13.8	0.0461	0.117	0.11	0.0023	—	2.9	-0.08
He 1 bar $Re=10000$	10000	3.18	2.99e+03	21.8	6.51e+04	45.57	50.3	54.3	46.4	5.00	125	26.8	0.0382	0.114	0.107	0.0003	—	5.6	-0.18
He 5 bar $Re=2300$	2300	0.731	140	0.395	55.5	47.47	67.1	79.8	46.6	20.82	30.1	6.45	0.0641	0.119	0.112	0.56	—	—	-0.04
He 18 bar $Re=2300$	2300	0.731	39	0.11	4.29	47.47	67.1	79.7	46.6	20.82	30.1	6.45	0.0643	0.119	0.112	7.3	—	—	-0.04
LN_2 3 bar $Re=500$	500	4.09	5.06	0.202	1.02	78.13	85.0	89.2	77.7	7.46	84	5.77	0.163	0.151	0.14	40	4.8	—	0.00
LN_2 3 bar $Re=500$ (g on)	500	4.09	5.06	1.68	8.5	77.32	78.9	81.7	77.7	1.74	361	24.8	1.35	0.65	0.478	9.4	12.3	—	-71.96
LN_2 3 bar $Re=1000$	1000	8.17	10.1	0.501	5.07	77.56	83.1	86.6	77.6	5.77	109	7.45	0.101	0.0977	0.0931	7.8	7.4	—	0.00
LN_2 3 bar $Re=1500$	1500	12.3	15.2	0.869	13.2	77.38	82.1	85.3	77.5	4.92	127	8.75	0.0778	0.0765	0.0736	2.9	8.7	—	0.00
LN_2 3 bar $Re=2300$	2300	18.8	23.3	1.57	36.6	77.25	81.2	84.1	77.5	4.12	152	10.4	0.0599	0.0595	0.0578	1	9.9	—	-0.00
LN_2 3 bar $Re=2300$ (g on)	2300	18.8	23.3	3	69.8	77.22	78.5	79.9	77.5	1.35	463	31.8	0.114	0.181	0.166	0.34	14.1	—	-10.77
LN_2 3 bar $Re=2300$ (coarse grid)	2300	18.8	23.3	1.55	36.1	77.25	81.2	84.1	77.7	4.09	153	10.5	0.0592	0.06	0.0582	1	9.9	—	0.01
LN_2 3 bar $Re=2300$ (fine grid)	2300	18.8	23.3	1.58	36.8	77.25	81.3	84.1	77.4	4.14	151	10.4	0.0603	0.0592	0.0575	1.1	9.9	—	-0.02
LN_2 3 bar $Re=5000$	5000	40.9	50.6	5.57	282	77.11	78.6	80.0	77.4	1.55	405	27.8	0.0449	0.073	0.0704	0.083	14.0	2.9	0.04
LN_2 3 bar $Re=10000$	10000	81.7	101	18.6	1.88e+03	77.06	77.8	78.6	77.4	0.81	777	53.3	0.0374	0.0699	0.0675	0.011	15.4	5.6	0.03

Table S8. Grid convergence index study for both fluids at $Re = 2300$ (Celik et al. 2008); meshes 86 400 / 291 600 / 984 150 cells, refinement ratio 1.5, production mesh = medium.

Fluid	Quantity	Coarse	Medium	Fine	p	Richardson	GCI_{medium}	GCI_{fine}
He, 1 bar	Nu	6.439	6.449	6.457	0.64	6.485	0.70 %	0.54 %
He, 1 bar	f	0.0632	0.0642	0.0647	1.40	0.0655	2.5 %	1.4 %
He, 1 bar	h [$\text{W m}^{-2}\text{K}^{-1}$]	30.05	30.10	30.14	0.64	30.27	0.70 %	0.54 %
LN ₂ , 3 bar	Nu	10.513	10.436	10.382	0.87	10.26	2.2 %	1.5 %
LN ₂ , 3 bar	f	0.0592	0.0599	0.0603	1.71	0.0607	1.6 %	0.8 %
LN ₂ , 3 bar	h [$\text{W m}^{-2}\text{K}^{-1}$]	153.2	152.1	151.3	0.87	149.5	2.2 %	1.5 %

Property correction and energy accounting. The helium transport properties of the earlier version of this comparison ($\mu = 2.73 \mu\text{Pa s}$, $k = 0.0185 \text{ W m}^{-1}\text{K}^{-1}$) correspond to helium near 15 K; the present runs use the 50 K values ($6.36 \mu\text{Pa s}$, $0.0467 \text{ W m}^{-1}\text{K}^{-1}$, CoolProp/NIST). The mixing-cup outlet temperature is the mass-flux-weighted average $\sum \phi_f T_f / \sum \phi_f$ over the outlet faces (`surfaceFieldValue`, `weightedAverage` with `phi`); the `weightedAreaAverage` operation, which multiplies the mass flux by the face area a second time, biases the outlet temperature low on a graded mesh and was the cause of the 2–10 % energy imbalances reported earlier. The delivered wall heat flux was verified with the solver-native `wallHeatFlux` function object (9.395 W).

Wall temperatures and the nitrogen loop pressure. Wall temperatures are read from the heated-wall patch with `surfaceFieldValue` (`max`, `min`, `areaAverage`; include file `system/wallTstats` in every case), and the wall-profile figure of the manuscript uses the patch face values along the centre row of the first channel (`postProcessing/wallFaces/wall_centreline.csv`). A line sample 0.4 mm inside the fluid, which an earlier post-processing used, underestimates the hottest wall face by 8–10 K for helium and 1–3.5 K for nitrogen and is not used anywhere in the paper. The nitrogen cases are stored under the name `LN2_3bar_*` because their properties were generated at 3 bar; the Boussinesq fields do not depend on the loop pressure (density, viscosity and conductivity of the 77 K liquid change by 0.06, 0.3 and 0.1 % between 3 and 5 bar), and the saturation margins of Table S7 are evaluated at the 5 bar loop pressure of the manuscript, with the 3 bar value (87.9 K) quoted where it matters.

S3 Hardware efficiencies used in the real ledger

Table S9. Stage efficiencies and the nitrogen supply-energy multiplier used in the real ledger.

Stage	η range	Source
4 K cryocooler stage	0.008–0.015	PT415: 1.5 W at 4.2 K with 40 W at 45 K for 10.7 kW [2]: the two stages together return 0.031 of their combined Carnot work, which the per-stage ranges reproduce to within 7 % at their midpoints; Radebaugh: “at 4 K efficiencies of about 1 % of Carnot or less are typical” [3]
He first stage (45–80 K)	0.10–0.15	Radebaugh: 10–15 % of Carnot for the best commercial 80 K coolers; large plants up to 30 % [3, 4]
LN ₂ supply (basis of Table 6)	$M_{N_2} = 9.1\text{--}12.7$ W per W intercepted	plant consumption 0.5–0.7 kWh kg ^{−1} [5] divided by the latent heat 199 kJ kg ^{−1} at 1 bar; equivalent to 0.30–0.40 of Carnot at 77 K (ideal liquefaction work 0.213 kWh kg ^{−1} [6]); the fraction-of-Carnot form gives 7.2–9.7 W/W and changes the nitrogen totals by 1–2 %

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